

Machine Learning Analysis of PCOS

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INTRODUCTION

Polycystic ovary syndrome (PCOS) is a complex endocrine and metabolic disorder, typically characterized by anovulation, infertility, obesity, insulin resistance, and polycystic ovaries. Lifestyle or diet, environmental pollutants, genetics, gut dysbiosis, neuroendocrine alterations, and obesity are among the risk factors that predispose females to PCOS. This project applies a complete data analysis pipeline to study PCOS using clinical, hormonal, and lifestyle variables. The workflow includes data preprocessing, feature selection, dimensionality reduction, and supervised learning. In addition, unsupervised clustering is used to explore hidden structures within the data. The goal is not only to build predictive models, but also to understand how different feature representations influence model performance and interpretability.

Dataset Exploration

The dataset selected for this project contains clinical, hormonal, and lifestyle information relevant to the study of PCOS. It was chosen because it includes a clearly defined target variable and a diverse set of features, making it suitable for both supervised classification and exploratory analysis.

- **Source:** Publicly available PCOS clinical dataset
- **Target variable:** PCOS (Y/N)

Feature characteristics:

- **Total number of variables:** Multiple features describing patient profiles
- **Numerical variables:** Hormonal measurements, follicle counts, anthropometric and clinical values
- **Binary variables:** Presence or absence of symptoms and lifestyle indicators (e.g., weight gain, hair growth, fast food consumption)
- **Categorical variables:** Limited and mostly encoded as binary indicators

At the initial stage of exploration, a small number of irrelevant columns, namely the patient identifier and patient file number, were identified. These variables do not carry predictive information and were therefore removed before proceeding with the subsequent data cleaning and preprocessing steps.

Missing Values

After identifying missing values, it was observed that they were limited to a few specific variables, namely **Marital Status, Fast Food consumption, and AMH (Anti-Müllerian Hormone)**.

The proportion of missing data was very small (approximately 0.184%, corresponding to a single row in the dataset). Given this negligible percentage, the affected row was safely removed from the dataset.

This approach avoided introducing any imputation bias while preserving the overall integrity and representativeness of the data for subsequent analysis.

Variables Encoding

After handling missing values, the dataset was examined to determine whether categorical variables required encoding. Several variables in the dataset represented binary responses in a **Yes/No** format, such as **PCOS status, pregnancy, weight gain, and lifestyle indicators**. Although these variables were conceptually categorical, they were already suitable for numerical analysis once converted into binary numeric form.

To ensure consistency and compatibility with machine learning algorithms, all Yes/No variables were explicitly converted to numeric values (0/1). This step guaranteed a homogeneous numerical representation across the dataset and prevented potential issues during later stages such as scaling, dimensionality reduction, and model training.

No additional encoding techniques (e.g., one-hot encoding) were required, as all categorical information was fully captured by these binary indicators.

Outliers Handling

After encoding, the dataset was examined for the presence of outliers that could negatively affect model performance. Outlier detection was carried out using boxplots and the interquartile range (IQR) method. This analysis was applied only to continuous numerical variables, as the concept of outliers is not meaningful for binary variables.

Binary indicators (Yes/No features encoded as 0/1) were intentionally excluded from outlier treatment to avoid distorting their semantic meaning.

For continuous variables, extreme values beyond the IQR bounds were identified and treated in order to reduce their influence while preserving the overall structure of the data. This step helped stabilize subsequent operations such as feature scaling, dimensionality reduction, and supervised learning, ensuring that models were not overly influenced by extreme observations.

Correlation Analysis

Following outlier handling, a correlation analysis was conducted to explore relationships between numerical variables and to identify potential redundancy within the feature set. Pearson correlation coefficients were computed for all numerical features, including the target variable, to assess the strength and direction of linear relationships.

This analysis served two main purposes. First, it helped identify variables that were strongly correlated with the target, making them potentially informative for prediction. Second, it revealed pairs of highly correlated features, indicating redundancy and the presence of multicollinearity.

These observations motivated the subsequent feature selection and dimensionality reduction steps. It is important to note that while correlation provides useful insights into linear dependencies, it does not capture non-linear relationships or interactions between variables and therefore cannot be used as the sole criterion for feature selection.

Feature selection

Based on the results of the correlation analysis, a correlation-based feature selection step was applied. Variables showing a meaningful correlation with the target variable were retained, while features with very weak associations were considered less informative and excluded. In addition, highly correlated feature pairs were examined to identify redundancy; when two variables conveyed similar information, only one was kept to reduce multicollinearity.

This step aimed to simplify the feature space while preserving the most relevant information for prediction. By reducing redundancy and noise, correlation-based feature selection improved model interpretability and provided a cleaner input for subsequent stages such as dimensionality reduction and supervised learning.

However, this selection was treated as an initial filtering step, recognizing that correlation alone does not account for non-linear relationships or feature interactions.

Feature Scaling

After feature selection, feature scaling was applied to ensure that all numerical variables contributed equally to subsequent analyses. Since the dataset contains variables measured on different scales (e.g., hormone levels, follicle counts, and physical measurements), standardization was necessary to prevent features with larger numeric ranges from dominating the learning process.

Standardization was performed using a z-score normalization approach, where each feature was transformed to have zero mean and unit variance. The scaler was fitted on the training data only and then applied to the validation and test sets to avoid data leakage. This step was essential for scale-sensitive methods such as PCA, SVM, and clustering, and it helped improve model stability and comparability across different feature representations.

Dimensionality Reduction

After completing the data preprocessing steps, dimensionality reduction was introduced as a means to further improve model performance and stability. The dataset contained several correlated variables, which can negatively affect learning algorithms by introducing redundancy and multicollinearity. To address this issue and to evaluate the impact of feature representation on predictive performance, dimensionality reduction techniques were applied before training the supervised models.

Two complementary approaches were considered. Principal Component Analysis (PCA) was used to reduce the feature space by projecting the data onto a smaller set of uncorrelated components while preserving most of the original information. The number of components was selected based on cumulative explained variance, retaining approximately 80–90% of the total variance.

This transformation aimed to provide a compact and efficient representation that could potentially enhance the performance of scalesensitive models.

In addition, Factor Analysis (FA) was applied to derive a lower dimensional representation based on latent variables underlying the observed features. The number of factors was selected using the Kaiser criterion (eigenvalues greater than one), and factor loadings were analyzed to ensure interpretability. The analysis revealed two dominant latent factors influencing PCOS: **ovarian morphology and insulin resistancerelated characteristics**. While FA was not primarily intended to maximize predictive accuracy, it provided an interpretable alternative feature representation that could be systematically compared with PCA and the original feature space.

Supervised Learning Models

After completing all preprocessing and representation steps, supervised learning models were trained to evaluate the predictive performance of the dataset under different feature representations. The objective was to assess how each preprocessing choice particularly dimensionality reduction impacts classification results.

Three supervised algorithms were considered: Logistic Regression, Support Vector Machine (SVM), and Random Forest. These models were selected because they represent different learning principles: linear modeling, margin-based classification, and ensemble-based decision trees. To ensure a fair comparison, the same train–test split, evaluation metrics, and preprocessing pipeline were used for all experiments.

Model training was conducted in a progressive manner. First, each algorithm was trained on the original preprocessed dataset (after cleaning, encoding, feature selection, and scaling) to establish a baseline performance. The same models were then retrained using the PCA transformed data and subsequently using the FA based representation. This experimental setup allowed a direct comparison of model performance across different feature spaces, isolating the effect of dimensionality reduction.

Performance was evaluated using appropriate classification metrics, with particular emphasis on the F1-score, as it provides a balanced measure of precision and recall. The results of these experiments are discussed in the following section, where the impact of feature representation on model accuracy and stability is analyzed.

Cross-Validation and Model Selection

To complement the single train–test evaluation and to obtain a more reliable assessment of model performance, cross-validation was applied. The goal of this step was to evaluate not only the average predictive performance of each model, but also its stability across different data splits.

K-fold cross-validation was performed on the training data, using a consistent number of folds for all models and feature representations. This approach allowed each model to be trained and validated multiple times on different subsets of the data. For each configuration, the mean performance and the variability across folds were computed using the same evaluation metric.

Cross-validation results highlighted important differences between models. In some cases, models that achieved relatively high performance on the test set exhibited noticeable variability across folds, indicating sensitivity to data partitioning. Conversely, models with slightly lower average scores but more consistent cross-validation results were considered more reliable. This analysis emphasized that model selection should not be based solely on peak performance, but also on robustness and generalization capability.

The insights gained from cross-validation were used to guide model comparison and selection before moving to the final exploratory clustering analysis.

Clustering Analysis

In addition to supervised classification, an unsupervised clustering analysis was conducted to explore the intrinsic structure of the data without using the target labels. The purpose of this step was not prediction, but rather to identify potential natural groupings among individuals based on their clinical and lifestyle characteristics.

Clustering was performed on the reduced feature spaces obtained through PCA and Factor Analysis. Using these representations helped mitigate the effects of noise and multicollinearity and ensured that clustering was applied on compact and meaningful feature spaces. The number of clusters was determined using silhouette analysis, which evaluates how well data points fit within their assigned clusters compared to other clusters.

The clustering results revealed well-defined groups with distinct characteristics. When applied to the PCA representation, clusters reflected differences in overall feature patterns, while clustering in the FA space enabled more interpretable groupings based on the previously identified latent factors. These clusters could be associated with varying clinical profiles, such as differences in ovarian morphology and metabolic or lifestyle-related features.

Overall, the clustering analysis provided complementary insights to the supervised learning results. While classification focused on predicting PCOS status, clustering offered an exploratory perspective, highlighting meaningful subgroups within the population and reinforcing the value of dimensionality reduction for both predictive and interpretative purposes.

Comparison of Supervised Models

The three supervised learning models: Logistic Regression, Support Vector Machine (SVM), and Random Forest, exhibited different behaviors across feature representations due to their underlying learning mechanisms.

Logistic Regression

showed relatively stable performance across the original, PCA, and FA representations. As a linear model, it is sensitive to multicollinearity but benefits from well-scaled and cleaned features. PCA slightly reduced redundancy, leading to comparable or marginally improved stability, while FA introduced a more abstract representation that resulted in a small performance decrease.

Overall, its consistency reflects the model's robustness when the feature space is well prepared.

Support Vector Machine (SVM)

achieved the highest performance, particularly when trained on PCA transformed data. SVM benefits significantly from reduced dimensionality and decorrelated features, as PCA simplifies the feature space and improves class separability. However, performance decreased when using FA, likely because FA prioritizes interpretability over maximizing discriminative variance, which is crucial for margin-based classifiers like SVM.

Random Forest

demonstrated the least improvement from dimensionality reduction. As an ensemble of decision trees, it is inherently capable of handling correlated and high-dimensional features. Consequently, PCA and FA did not provide substantial benefits and, in some cases, led to reduced performance due to the loss of detailed feature information that Random Forest models typically exploit.

This comparison highlights that model performance is closely linked to the chosen feature representation. While PCA enhanced performance for models sensitive to feature correlation, such as SVM, FA offered interpretability at the cost of predictive power. These variations underscore the importance of aligning model choice with feature representation and analysis objectives.

Conclusion

This project applied a complete machine learning pipeline to the analysis of Polycystic Ovary Syndrome using clinical and lifestyle data. Starting from data cleaning and preprocessing, each step was carefully designed to improve data quality and ensure reliable modeling outcomes. Feature selection and scaling helped reduce noise and prepare the data for advanced analysis.

Dimensionality reduction played a central role in the study. PCA proved effective in reducing redundancy and improving model performance, particularly for models sensitive to feature correlation. Factor Analysis, while slightly less performant in some cases, provided valuable interpretability by revealing meaningful latent clinical structures. Comparing models across original, PCA-based, and FA-based representations highlighted the importance of feature representation in supervised learning.

Supervised models demonstrated varying behavior depending on the representation used, and cross-validation emphasized the need to consider performance stability in addition to raw accuracy. Finally, clustering analysis offered an exploratory perspective, revealing natural groupings within the data that complemented the classification results.

Overall, this study shows that combining thorough preprocessing, dimensionality reduction, supervised learning, and clustering leads to both improved predictive performance and deeper insight into the underlying structure of PCOS-related data.

Limitations and Future Work

Despite the strong performance and insights obtained, this study has several limitations. First, the dataset size is relatively limited, which may affect the generalization ability of the models when applied to unseen populations. In addition, correlation-based feature selection and linear dimensionality reduction techniques may not fully capture complex non-linear relationships present in medical data.

Future work could explore more advanced feature selection methods and non-linear dimensionality reduction techniques. Incorporating additional data sources, such as longitudinal patient records or genetic information, could further enhance predictive performance. Finally, evaluating the models on external datasets would provide a more robust assessment of their generalizability and practical applicability.

